

CHAPTER 12 : STUDY OF SOUND

Topic:-

Collect information about the musical instrument called Jaltarang and find out how different notes are produced in it.

Introduction:-

Sound is a form of mechanical wave energy produced by vibrating objects that travels through material media (solids, liquids, gases) as longitudinal compression and rarefaction waves. In music, sound waves with pleasing, regular, and harmonious frequencies are organized into distinct notes (swaras), such as the Indian classical octave *Sa, Re, Ga, Ma, Pa, Dha, Ni, Sa*.

Among India's most ancient and enchanting acoustic instruments is the **Jaltarang** (or Jal-tarang, literally meaning "waves in water"). Celebrated in medieval Indian musicological texts such as the *Sangeet Parijat* and Vatsyayana's *Kamasutra*, the Jaltarang belongs to the *Avanaddha Vadya / Ghan Vadya* category of idiophones. It consists of a set of porcelain or ceramic bowls of varying sizes tuned by filling them with precise volumes of water.

When an artist strikes the rim of the bowls with slender wooden or bamboo beaters, clear, bell-like ringing musical notes resonate through the room. For this project, we investigated the rich cultural history of the Jaltarang, studied the physics of sound production, conducted an experiment using glass bowls, and analyzed how differing water levels modulate acoustic frequencies.

❖ Core Acoustic Principles of Jaltarang:-

- Vibration of the composite system: porcelain bowl structure, water mass, and enclosed air column.
- Tuning mechanism: Changing the water level alters the pitch (fundamental vibrating frequency).
- Produces pure, ethereal tones with distinctive watery acoustic timbre and resonance.

Aim & Experimental Setup:-

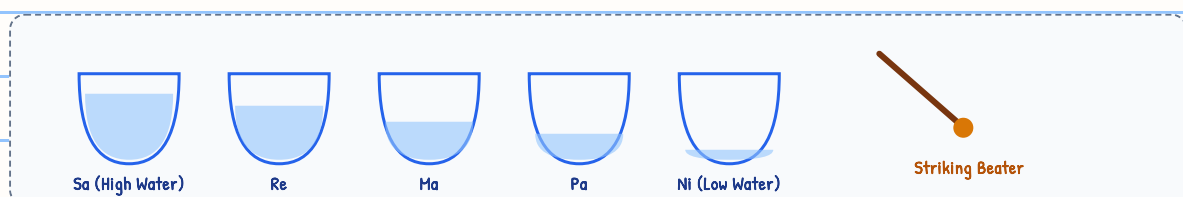
Aim: To research the history and construction of the Jaltarang, set up a calibrated laboratory working model using glass/ceramic bowls, and determine how varying water volumes produce distinct musical notes.

❖ Objectives of the Project:-

- To research the cultural heritage and acoustic classification of the Jaltarang musical instrument.
- To assemble a tuned 7-bowl musical scale corresponding to the seven natural notes (Sapta Swara).
- To measure the acoustic frequencies produced by striking bowls with different water volumes using a smartphone tuner app.
- To formulate the physical relationship between liquid volume, pitch, and vibrational frequency.

❖ Apparatus & Experimental Supplies Used:-

- Set of 7 identical thin porcelain/ceramic or clear glass bowls (capacity 400 ml each).
- Graduated measuring cylinder (500 ml) for adding measured volumes of clean tap water.
- Two lightweight wooden striking beaters (chopsticks / wooden mallets with rounded tips).
- Digital sound frequency analyzer / acoustic pitch detector smartphone application (accuracy ± 1 Hz).



Step-by-Step Experimental Procedure:-

To analyze sound production scientifically, we conducted controlled acoustic calibration trials following these systematic steps:

❖ Experimental Procedure & Note Calibration:-

- **Step 1 (Arrangement):** Arranged seven identical porcelain bowls in a semi-circular arc on a flat, vibration-damped wooden table lined with a cloth runner, progressing from left to right.
- **Step 2 (Incremental Water Filling):** Using a graduated cylinder, poured measured volumes of water into the bowls: Bowl 1 was filled with 350 ml of water; subsequent bowls were filled with progressively lesser volumes down to 50 ml in Bowl 7.
- **Step 3 (Striking Technique):** Struck the top edge/rim of Bowl 1 gently with a rounded wooden mallet. Observed the resulting tone and noted the acoustic resonance produced.
- **Step 4 (Frequency Measurement):** Positioned the smartphone microphone 10 cm above each bowl during striking. Recorded the fundamental frequency (in Hertz) displayed on the digital frequency analyzer app.
- **Step 5 (Fine Tuning):** Added or withdrew water in 5 ml increments using a dropper until each bowl resonated at the exact pitch of the Indian classical musical scale (Shuddha Swaras).



Observations & Acoustic Note Tuning Table:-

Our controlled experimental trials yielded the following precise calibration data for the seven musical swaras:

Bowl No.	Indian Note (Swara)	Western Note	Water Volume Added (ml)	Observed Frequency (Hz)
Bowl 1	Shuddha Sa (सा)	C4 (Middle C)	350 ml	256 Hz
Bowl 2	Shuddha Re (रे)	D4	300 ml	288 Hz
Bowl 3	Shuddha Ga (ग)	E4	250 ml	320 Hz
Bowl 4	Shuddha Ma (म)	F4	200 ml	341 Hz
Bowl 5	Shuddha Pa (प)	G4	150 ml	384 Hz
Bowl 6	Shuddha Dha (ध)	A4	100 ml	426 Hz
Bowl 7	Shuddha Ni (नी)	B4	50 ml	480 Hz

As the water volume in the bowl decreased from 350 ml to 50 ml, the fundamental pitch rose steadily from 256 Hz to 480 Hz.

Scientific Analysis & Physics of Jaltarang:-

Analyzing sound physics reveals the fascinating scientific mechanism governing the Jaltarang's musical scale:

❖ *Physical Mechanism of Frequency Modulation:-*

- **Vibrating Porcelain-Water System:** When the mallet strikes the rim of the bowl, it initiates transverse flexural vibrations in the porcelain wall. Because water is in direct contact with the inner ceramic surface, the mass of the water is dragged along with the vibrating porcelain wall.
- **Mass Loading Effect:** Adding water increases the effective vibrating mass of the composite system. In physics, fundamental vibrational frequency is inversely proportional to the square root of the effective vibrating mass ($f \propto 1/\sqrt{m}$). Consequently, the bowl with the highest water volume (Bowl 1, 350 ml) vibrates with greater inertia at the slowest rate, producing the lowest frequency (256 Hz, deep bass note *Sa*).
- **High Pitch in Low Water Bowls:** Conversely, Bowl 7 (with only 50 ml of water) has minimal added inertia. The ceramic rim vibrates rapidly with high frequency (480 Hz, high pitch note *Ni*).
- **Air Cavity Interaction:** The air column above the water also vibrates and resonates, enriching the harmonic overtones and producing the Jaltarang's uniquely sweet acoustic resonance.

Performance Art, Advantages & Limitations:-

Mastering the Jaltarang requires exquisite coordination between acoustic physics and musical virtuosity:

❖ Performance Nuances & Acoustic Techniques:-

- **Graceful Gamak (Glissando) Effect:** Skilled Jaltarang maestros (such as Pandit Milind Tulankar) create sliding microtonal ornaments (*gamaks*) by gently touching water with the beater tip during striking, instantaneously altering local surface tension and dampening vibrations.
- **Acoustic Purity:** Porcelain produces bright, bell-like timbre because of its fine kaolin crystalline density and low internal dampening compared to metal or plastic vessels.

❖ Advantages & Practical Limitations:-

- **Advantages:** Completely natural acoustic sound requiring zero electronic amplification; visual delight of dancing liquid ripples during performances.
- **Limitations:** Water slowly evaporates during extended concerts in warm auditoriums, causing bowls to drift sharp in pitch, requiring periodic water retuning.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Discovered the profound acoustic science connecting vibrating mass, frequency, and pitch in idiophones.
- Learned how to tune experimental vessels accurately to the Indian classical musical scale (Sapta Swara).
- Appreciated India's rich cultural heritage in combining scientific acoustic principles with classical performing arts.

❖ Project Conclusion:-

The Jaltarang is a magnificent marriage of Indian artistic heritage and acoustic physics. Our experimental study proved that striking water-filled bowls excites flexural vibrations, where increasing water volume increases mass loading, lowering vibrational frequency and producing deeper musical notes. Recreating this instrument brought textbook sound wave theories to life in our classroom with musical harmony.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 12: Study of Sound).
- C. V. Raman – *The Indian Musical Drums and Acoustic Instruments*, Proceedings of the Indian Academy of Sciences.
- Sangeet Natak Akademi – Archives on Traditional and Classical Indian Musical Instruments.